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Thermophoresis of spherical and non-spherical particles: a review of theories and experiments

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Abstract

Thermophoresis is an important mechanism of micro-particle transport due to a temperature gradient in the surrounding medium and has found numerous applications, especially in the field of aerosol technology. Extensive studies, both theoretical and experimental, have been done to understand the nature of this phenomenon. However, it is clear that a lot more of work needs to be done before we can predict thermophoresis accurately for any given gas-particle system as well as particle shape and orientation in any flow regime. This paper reviews the existing theories and data in two major categories, for spherical particles and for non-spherical particles, as well as the various techniques in making thermophoresis measurements. The current state of development for thermophoresis studies is that for spheres the theories and experimental data agree with each other fairly well but for non-spherical particles in the transition regime the theories are yet to be developed and experimental data showing the effect of particle shape are much needed in all Knudsen number range. The best techniques of thermophoretic force measurements involve the use of electrodynamic balances to work on single micro-particles and the use of microgravity to minimize the effect of convection. A combination of the above two has not been attempted and should provide the most accurate data. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Thermophoresis; Thermophoretic force; Non-spherical particle

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1. Introduction

In 1870 J. Tyndall [1] observed that in a chamber filled with dust-laden air and illuminated properly, particle-free regions formed around a hot metal ball or a heated platinum wire, which appeared as a dark space in the bright background of scattered light by the dust particles. A decade later, this interesting phenomenon was examined in closer detail by Rayleigh [2] and Lodge and Clark [3], but it remained unclear why the dust-free region should form. It was Aitken [4] who proved, with a series of experiments in 1884, that the dust particles must have been driven away from the heated surface by differential bombardment of the gas molecules. The molecules moving from the hot surface carried greater kinetic energy than those moving from the cold side, resulting in a net force on the particle. This was indeed the correct picture at the microscopic level, as illustrated in Fig. 1, and the resulting net force is called *thermophoretic force* (TF). The phenomenon in which particles migrate in a temperature gradient under the influence of the thermophoretic force is called *thermophoresis*, and the resulting migration velocity of the particles is called *thermophoretic velocity*.

Over more than 100 years since its discovery, the importance of thermophoresis has been recognized in numerous applications where it can play either beneficial or adverse roles. Thermophoresis has been exploited in the design of thermal precipitators to remove small particles from gas streams. Aerosol sampling methods based on the principle of thermophoresis have been used extensively in combustion related studies. It has been found that thermophoresis caused particles to deposit on boiler pipes, reducing the efficiency of heat exchange. The effect of thermophoresis on the transfer of the radioactive aerosols generated in a nuclear reactor accident has been recognized as an important factor in reactor safety assessment. In the semiconductor industry thermophoresis has been used as an effective method of microcontamination control. The wafers are heated so that the

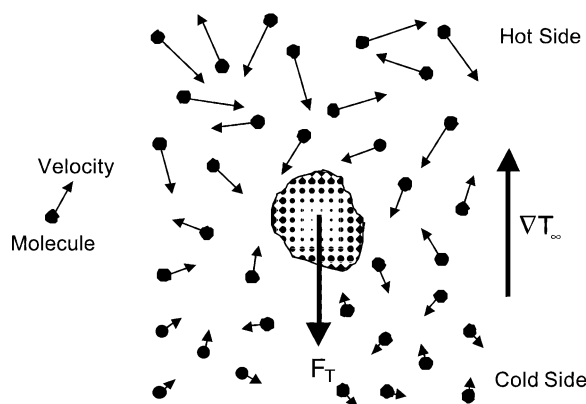


Fig. 1. A particle immersed in a gas with a temperature gradient.

thermophoretic force repels undesired particles from the surface, preventing their deposition and subsequent damage to the chips. However, thermophoresis can be used to enhance chemical vapor deposition processes. Fine particles can be produced by gas phase reactions and deposited on a substrate of lower temperature to form a thin film. This technology has been used in the fabrication of optical fibers. In an environment under microgravity conditions, such as that on a spacecraft orbiting the earth, thermophoresis can be the dominant mechanism of fine particle transfer. This has important implications in the event of a spacecraft fire accident and also in future microgravity manufacturing processes.

Because of the practical importance of thermophoresis, it has been the subject of extensive study over the years. On the theoretical side, it is clear that the ultimate challenge is to solve the Boltzmann equation for the molecular velocity distribution function for a gas with a temperature gradient. With the knowledge of the velocity distribution function, the thermophoretic force can be calculated by integration over the entire particle surface for the net momentum transfer from the gas molecules. However, the highly non-linear nature of the Boltzmann equation prohibits its exact solution at present.

Under certain conditions, however, approximations can be made to greatly reduce the above difficulty. Experimental data have shown that thermophoresis strongly depends on the Knudsen number, or the ratio of the gas mean free path ℓ to a characteristic length scale of the particle,

$$Kn = \frac{\ell}{a} \quad (1)$$

Here the appropriate particle length scale is assumed to be the equivalent-volume radius a of the particle. In the so-called *near-continuum regime* ($Kn \rightarrow 0$), the gas is not far from a state of a continuous medium, while in the *free-molecule regime* ($Kn \rightarrow \infty$), the role of the particle resembles that of a very large molecule. In both

of the above two limits, the derivation of the thermophoresis theory can be simplified, while in the *transition* or *Knudsen regime* of intermediate Knudsen number, the theoretical solutions are the most difficult to obtain.

Historically, there have been three theoretical approaches in thermophoresis study that correspond to the above division of the Knudsen number range. The first one, pioneered by Epstein [5] in 1929, is based on the theory of continuum fluid mechanics, in which the governing equations are the special case of the Boltzmann equation in the limit of $Kn \rightarrow 0$. The key here is the use of slip flow boundary conditions to construct a thermophoresis theory that is correct in the near-continuum regime. This approach was refined by other researchers to incorporate a full set of slip boundary conditions [6] and to include higher order terms [7], which extended the solutions to the near-continuum regime of higher Knudsen number.

The second approach is applicable in the limit of $Kn \rightarrow \infty$. Under such a condition, the velocity distribution of the gas molecules is not disturbed much by the presence of a particle, which greatly simplifies the mathematics in thermophoretic force calculation. The velocity distribution function in the absence of the particle can be obtained by the Chapman–Enskog method, which involves a power series expansion around the known solution for the Boltzmann equation in equilibrium state, i.e. the Maxwellian distribution function. Using this approach, Waldmann [8] first established the free molecule limit of thermophoretic force in 1959. Others attempted to extend this approach to smaller Knudsen number but only with limited success.

The third approach of the theoretical thermophoresis study is to solve the Boltzmann equation numerically, which only became viable because of the advances in computers in recent years. Solving the Boltzmann equation in its full form is still a formidable task, however. Instead, the Boltzmann equation or its various approximate models was linearized around the equilibrium state and solved numerically. This approach, first used by Sone and Aoki [9] in 1983, was adopted by several groups of authors to calculate the thermophoretic force in the entire range of Knudsen numbers, including the transition regime.

There has also been significant development in the experimental aspect of thermophoresis research. From the measurements of dark space thickness in the early years to the recent experiments on single particles and those carried out under microgravity conditions, a large number of data have been accumulated.

It should be emphasized that the vast majority of the previous results, both theoretical and experimental, are for the thermophoresis of spheres. Theoretical development for the non-spherical particle thermophoresis is scanty and the only few results available are the near-continuum and the free molecule asymptotic solutions, which indicate that the effect of particle shape and orientation is significant. Experimental data for non-spherical particle thermophoresis have been very limited. Given the fact that most solid aerosols of practical interests are rarely spheres, this large gap in our knowledge in thermophoresis must be bridged with more data and better theories. In this review the existing thermophoresis theories and data for both spherical and non-spherical particles as well as the various

measurement techniques for thermophoresis found in the literature are summarized.

2. Previous results for spherical particles

In this section the previous theoretical and experimental results for the thermophoresis of spherical particles are given in detail. Note that many authors' definitions of the mean free path were based on the gas thermal conductivity κ_2

$$\ell = \frac{4}{5} \frac{\kappa_2}{p} \sqrt{\frac{mT}{2k_B}} \quad (2)$$

where p and T are the pressure and the temperature of the gas, respectively, m is the mass of a molecule and k_B is the Boltzmann constant.

The alternate form of the above definition based on the gas viscosity is

$$\ell = \frac{\mu_2}{p} \sqrt{\frac{2k_B T}{m}} \quad (3)$$

where the viscosity μ_2 is related to the thermal conductivity by the Bhatnagar–Gross–Krook (BGK) model of the Boltzmann equation,

$$\mu_2 = \frac{2}{5} \frac{m}{k_B} \kappa_2 \quad (4)$$

Some authors' definitions of the mean free path were different from the above definitions by a constant factor

$$\ell^* = \frac{\sqrt{\pi}}{2} \ell \quad (5)$$

In this paper, the definition based on the gas thermal conductivity is used for the mean free path, as given by Eq. (2). Other definitions found in the literature were all converted to this form. For polyatomic gases the contribution to the thermal conductivity due to the translational motion of the gas molecules, κ_{tr} , is used to calculate the mean free path,

$$\kappa_{tr} = \frac{15}{4} \frac{k_B}{m} \mu_2 \quad (6)$$

2.1. Free molecule regime

In the limit of $Kn \rightarrow \infty$ the velocity distribution function of the incoming molecules is not affected by the presence of the particle. The velocity distribution

function, as mentioned above, can be obtained by the Chapman–Enskog method for a gas in non-equilibrium state. To calculate the thermophoretic force, it is necessary, however, to model the manner in which the gas molecules are reflected by the particle surface after collisions.

In the *specular reflection* model, the collision of the gas molecules with the particle surface is completely elastic. In the *diffuse reflection* model, gas molecules colliding with the particle become in equilibrium with the particle surface and leave the surface with a Maxwellian distribution of velocity specified at the temperature of the particle surface. The way in which gas molecules actually collide with the particle is most probably something between the above two extremes. This is conventionally modeled by the introduction of *accommodation coefficients*.

For example, the *momentum accommodation coefficient* is defined by

$$\alpha_m = \frac{M_i - M_e}{M_i - M_s} \quad (7)$$

where M is the molecular momentum averaged over the entire velocity space, and the subscripts i, e and s denote molecules emerging from, impinging on the surface, and molecules leaving the surface with a Maxwellian velocity distribution at the surface temperature, respectively. Similarly, the *thermal accommodation coefficient* is defined by

$$\alpha_t = \frac{E_i - E_e}{E_i - E_s} \quad (8)$$

where E is the average energy flux. Accommodation coefficients of value one signify complete accommodation of the gas molecules. For diffuse reflection of the gas molecules $\alpha_m = \alpha_t = 1$ and for specular reflection $\alpha_m = \alpha_t = 0$.

Waldmann [8] was the first to derive the free-molecule limit of the thermophoretic force for monatomic gases,

$$F_{T,FM} = -\frac{16\sqrt{\pi}}{15} \frac{a^2 \kappa_2}{\sqrt{2k_B T_0/m}} \nabla T_\infty \quad (9)$$

where T_0 is the average gas temperature at the particle, ∇T_∞ is the temperature gradient far away from the particle. Note that Waldmann's free molecule limit is proportional to the square of the particle radius and independent of its thermal conductivity. After Waldmann's work was published, Mason and Chapman [10] showed that the free-molecule thermophoretic force could also be obtained from the dusty-gas model. Their result was different from Waldmann's by a factor of constant because they only considered elastic collisions. However, Monchick et al. [11] later showed that the dusty-gas model agreed with Waldmann's result if inelastic collision was accounted for.

It is a surprise to see that the free molecule limit of thermophoretic force does not depend on the accommodation coefficients. Phillips [12] derived the following

equation that allowed the treatment of the molecular scattering at the particle surface in more detail,

$$F_{T,FM,Phillips} = F_{T,FM} \left(1 + \frac{1}{2}(\alpha_{mt} - \alpha_{mn}) \right) \quad (10)$$

where α_{mt} and α_{mn} are the tangential and normal momentum accommodation coefficients, respectively, and are defined by Eq. (7) with the molecular momentum terms replaced by the corresponding components. Experimental data show that most gas–solid interfaces have α_{mt} close to unity. The normal momentum accommodation coefficient is relatively poorly known but observations are consistent with $\alpha_{mn} \approx 1$. Therefore, Waldmann's free molecule limit is more accurate than for which it was originally derived.

Although the above free molecule limits are derived for monatomic gases, they can be extended to polyatomic gases by replacing the gas thermal conductivity with its translational part given by Eq. (6). Experiments over the years [13–20] have showed that in the free molecule regime the thermophoretic force indeed approaches Waldmann's limit as the particle Knudsen number increases.

It is convenient to define a dimensionless thermophoretic force for particles of arbitrary shape based on the dependence on temperature, gas thermal conductivity, temperature gradient and particle size indicated in the above free molecule limit for spheres:

$$f_T = -F_T \left(\frac{a^2 \kappa_2}{\sqrt{2k_B T_0/m}} \nabla T_\infty \right)^{-1} \quad (11)$$

Here a is of course the radius for a sphere. It is assumed that for a non-spherical particle the appropriate length scale a for normalization is the equivalent-volume radius. The Waldmann free molecule limit of the thermophoretic force for spheres then becomes

$$f_{T,FM} = \frac{16\sqrt{\pi}}{15} \approx 1.8906 \quad (12)$$

2.2. Near-continuum regime

In the limit of $Kn \rightarrow 0$ the Boltzmann equation can be shown to reduce to the Navier–Stokes equations of continuum fluid mechanics. The thermophoresis theory in the *near-continuum regime* ($Kn \ll 1$) can be built upon the solution of the Navier–Stokes equations combined with the appropriate slip boundary conditions. The latter accounts for the fact that the gas cannot be treated as a continuous medium within a few mean free paths from the particle surface (the *Knudsen layer*).

Epstein [5] developed a solution in this manner, but the slip conditions that he used were not complete. Therefore, his result agreed with the data for oil droplets

of relatively low thermal conductivity by Rosenblatt and LaMer [21] and Saxton and Ranz [22], but failed to correlate the thermophoretic force data for high thermal conductivity particles (sodium chloride particles and mercury droplets) by Schadt and Cadle [20]. Brock [6] was the first to derive a near-continuum solution with the complete set of slip conditions. His equation for thermophoretic force in dimensionless form is

$$f_T = \frac{24\pi}{5} \frac{C_{tc} Kn(\kappa_{21} + C_t Kn)}{(1 + 3C_m Kn)(1 + 2\kappa_{21} + 2C_t Kn)} \quad (13)$$

where κ_{21} is the ratio of the gas and the particle thermal conductivities. The velocity slip coefficient C_m , the temperature jump coefficient C_t and the thermal creep coefficient C_{tc} are defined in the first order slip boundary conditions. They depend on the intermolecular force law and the molecular scattering model at the particle surface. These can be calculated from gas kinetic theory and also measured.

As an asymptotic solution for the thermophoretic force in small Knudsen number to the first order, Brock's solution failed to match the data of Jacobson and Brock [18] at larger Knudsen numbers for sodium chloride aerosols. In an attempt to obtain a higher order solution based on the linearized 13-moment theory, Dwyer [23] first noted the possibility that the thermophoretic force for high thermal conductivity particles at small Knudsen numbers might reverse its normal direction and become in the same direction as the temperature gradient. This phenomenon was termed *negative thermophoresis* later by Sone [24].

Sone and Aoki [7,25] derived an asymptotic solution for the thermophoretic force up to the third order in small Knudsen number based on the Boltzmann–Krook–Wendland (BKW) model of the Boltzmann equation. This solution exhibits negative thermophoresis for infinite particle thermal conductivity ($\kappa_{21} = 0$) at small Knudsen numbers. Later, they derived a solution [9] of the BKW equation for $\kappa_{21} = 0$ using a variational method. The new solution predicts that negative thermophoresis occurs for $Kn = 0$ –0.2, and the magnitude of the reversed thermal force is no more than 4% of Waldmann's free molecule limit.

More recent solutions of the BGK model of the Boltzmann equation [26,27] confirm the existence of a weak negative thermophoretic force for finite but large particle thermal conductivity ($\kappa_{21} = 0.01$). However, Loyalka [27] cautioned that negative thermophoresis might be an artifact due to the BGK approximation. Although he solved the linearized Boltzmann equation directly for hard sphere molecules, it was not clear whether the negative thermophoresis exists for the hard sphere model because his numerical computation became especially lengthy as Kn decreased, and he had to stop at $Kn = 1$. More recently, Takata et al. [28] were able to carry out the calculation for the hard sphere model down to $Kn \approx 0.096$ with the assumption of infinite particle thermal conductivity and a numerical scheme different from Loyalka's. However, no reversal in the direction of thermophoretic force was found. They did a preliminary numerical computation at

$Kn \approx 0.019$ and found a very small negative thermophoretic force but were not sure whether it was just a manifestation of the numerical error.

To summarize, it remains unclear whether negative thermophoresis is an artifact due to the models of the Boltzmann equation. There is no experimental evidence of negative thermophoresis. The magnitude of the thermophoretic force at small Knudsen numbers is very small and other forces such as the drag due to gas convection can be easily overwhelming. So far no attempt has been made to verify negative thermophoresis experimentally.

2.3. Transition regime

As mentioned above, there have been attempts to extend the asymptotic solutions near the continuum limit to the transition regime with limited success. Similar attempts had been made to extend the free molecule solution to finite but large Knudsen number region [29,30]. Talbot [31] recognized that in the limit of $Kn \rightarrow \infty$, Brock's near-continuum solution, Eq. (13), reduces to a form that differs from the free molecule solution of Waldmann, Eq. (9), only by a constant,

$$\lim_{Kn \rightarrow \infty} f_{T,Brock} = f_{T,FM,Waldmann} \left(\frac{3\sqrt{\pi}}{4} \frac{C_{tc}}{C_m} \right) \quad (14)$$

Using the best available data for C_{tc} and C_m , he found that the constant $\left(\frac{3\sqrt{\pi}}{4} \frac{C_{tc}}{C_m} \right)$ is only 3% larger than unity, and suggested the use of Brock's equation as an interpolation formula for the thermophoretic force in the entire range of Knudsen numbers. This formula was used widely before the more accurate solutions of the Boltzmann equation became available.

Sone and Aoki [9] applied a variational method to the BKW model of the linearized Boltzmann equation and obtained a similarity solution for the thermophoresis of a sphere of infinite thermal conductivity and diffuse reflection. Their results were presented graphically in the range $0 < Kn < 10$. Their solution for the thermophoretic force approaches Waldmann's free molecule limit $f_{T,FM} = 16\sqrt{\pi}/15$ as $Kn \rightarrow \infty$.

Yamamoto and Ishihara [26] solved the BGK model of the Boltzmann equation numerically for diffuse-reflected molecules and relaxed the assumption of infinite thermal conductivity of the sphere that Sone and Aoki made. Their solution for the thermophoretic force can be expressed in the dimensionless form

$$f_T = \frac{16\pi}{5} \left[A_w H_o - A_o \left(H_w + \frac{5\sqrt{\pi}}{4} \frac{Kn}{\kappa_{21}} \right) \right] \left(H_w + \frac{5\sqrt{\pi}}{4} \frac{Kn}{\kappa_{21}} \right)^{-1} \quad (15)$$

where A_w , A_o , H_w and H_o are functions of the Knudsen number and were tabulated for $0.01 \leq Kn \leq 50$ in their original paper. The values of these coefficients are included here in Table 1. This solution is particularly convenient for

Table 1

The values of the coefficients in Yamamoto and Ishihara's solution [26] for thermophoretic force

Kn	A_ω	A_o	H_ω	H_o
0.01	-0.3772×10^{-2}	0.9056×10^{-4}	0.04319	-0.06501
0.05	-0.01867	0.1849×10^{-2}	0.1918	-0.2933
0.1	-0.0352	0.4379×10^{-2}	0.3316	-0.5195
0.2	-0.05612	0.1741×10^{-2}	0.5117	-0.8472
0.3	-0.06469	-0.8786×10^{-2}	0.6185	-1.085
0.5	-0.06612	-0.03526	0.7364	-1.448
0.75	-0.05981	-0.06381	0.8104	-1.814
1	-0.05276	-0.08512	0.8521	-2.142
2	-0.03381	-0.1298	0.9215	-3.342
3	-0.02441	-0.1485	0.9469	-4.488
5	-0.0156	-0.1642	0.964	-6.719
7.5	-0.01065	-0.1716	0.976	-9.511
10	-0.8094×10^{-2}	-0.1759	0.9817	-12.29
20	-0.4113×10^{-2}	-0.1812	0.9902	-23.38
30	-0.2756×10^{-2}	-0.1833	0.9935	-34.47
50	-0.1659×10^{-2}	-0.1828	0.9961	-56.63

comparing different theories and experimental data because the thermophoretic force can be calculated for arbitrary values of gas thermal conductivity.

Loyalka [27] solved the linearized Boltzmann equation for hard sphere molecules and diffuse reflection by a finite element and collocation method, in which the first few terms of the associate Legendre polynomial expansion of the hard-sphere collision kernel were retained. He calculated the thermophoretic force for five values of the Knudsen number between 1 and 10 and two values of the thermal conductivity ratios. The numerical computation was especially lengthy, and he was not able to go lower than $Kn = 1$ due to computing hardware limitations. Loyalka also solved the linearized BGK model of the Boltzmann equation to verify the accuracy of his numerical scheme. His solutions for both the hard sphere gas and the BGK model are presented in Table 2.

Takata et al. [28] solved the linearized Boltzmann equation for hard sphere

Table 2

Dimensionless thermophoretic force calculated by Loyalka [27]

Kn	f_T (hard sphere gas)		Kn	f_T (BGK model)	
	$\kappa_{21} = 0.1$	$\kappa_{21} = 0.01$		$\kappa_{21} = 0.1$	$\kappa_{21} = 0.01$
1	0.8649	0.8232	0.1	0.03212	-0.04465
1.3...	1.0618	1.0292	0.2	0.09872	-0.0021
2	1.3167	1.292	1	0.9339	0.8869
4	1.6207	1.6092	2	1.3492	1.3222
10	1.8158	1.8142	10	1.78	1.776

Table 3
Solution for the dimensionless thermophoretic force by Takata et al. [28]

Kn	f_T
0.09611	0.007
0.1922	0.046
0.2883	0.115
0.3844	0.208
0.5767	0.412
0.7689	0.602
1.153	0.903
1.922	1.259
3.845	1.6
7.689	1.782
11.53	1.84
19.22	1.884
∞	1.942

molecules, diffuse reflection, and infinite particle thermal conductivity by a method combining the hybrid difference scheme and the numerical kernel method. They tabulated the dimensionless thermophoretic force in the range $0.096 \leq Kn \leq \infty$, as shown in Table 3.

All of the above numerical solutions for the Boltzmann equation approach

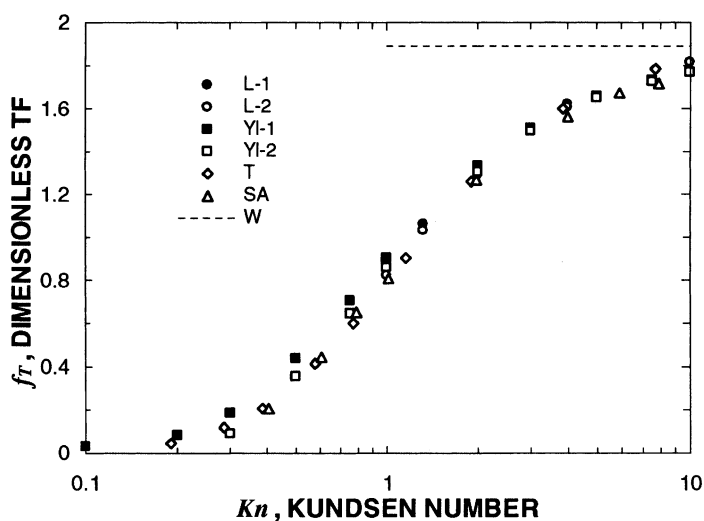


Fig. 2. Comparison of the dimensionless thermophoretic force predicted by the solutions of the Boltzmann equation. L-1: $\kappa_{21} = 0.1$, L-2: $\kappa_{21} = 0.01$, hard sphere gas, Loyalka [27]; YI-1: $\kappa_{21} = 0.1$, YI-2: $\kappa_{21} = 0$, BGK model, Yamamoto and Ishihara [26]; T: $\kappa_{21} = 0$, Takata et al. [28]; SA: $\kappa_{21} = 0$, Sone and Aoki [9]; W: free molecule limit, Waldmann [8].

Waldmann's free molecule solution $f_{T,FM} = 16\sqrt{\pi}/15$ in the limit of $Kn \rightarrow \infty$, except for the solution of Takata et al., which has a free molecule limit of 1.942. These theoretical solutions are compared in Fig. 2. Note that in the transition regime the agreement among the theories is good and the effect of thermal conductivity appears to be rather small.

The above solutions of the Boltzmann equations and the interpolation formula

Table 4

Thermal conductivity ratios of the data shown in Figs. 3 and 4

Legend	Particle	Gas	κ_{21}	Reference
RL	Tricresyl phosphate	Air	0.123	Rosenblatt and LaMer [21]
SR	Castor oil	Air	0.110	Saxton and Ranz [22]
S	Paraffin PH300	Argon	0.237×10^{-2}	Schmitt [19]
SC-1	Sodium chloride	Air	0.313×10^{-2}	Schadt and Cadle [20]
SC-2	Tricresyl phosphate	Air	0.101	Schadt and Cadle [20]
JB-1, JB-2	Sodium chloride	Argon	0.427×10^{-2}	Jacobsen and Brock [18]
KO-1	Magnesium oxide	Air	0.530×10^{-3}	Keng and Orr [32]
KO-2	Aluminum oxide	Air	0.596×10^{-3}	Keng and Orr [32]
LD-1	Nickel	Air	0.219×10^{-3}	Li and Davis [13]
LD-2	Nickel	Helium	0.165×10^{-2}	Li and Davis [14]
LD-3	Polystyrene latex	Air	0.164	Li and Davis [13]
LD-4	Polystyrene latex	Air	0.164	Li and Davis [13]

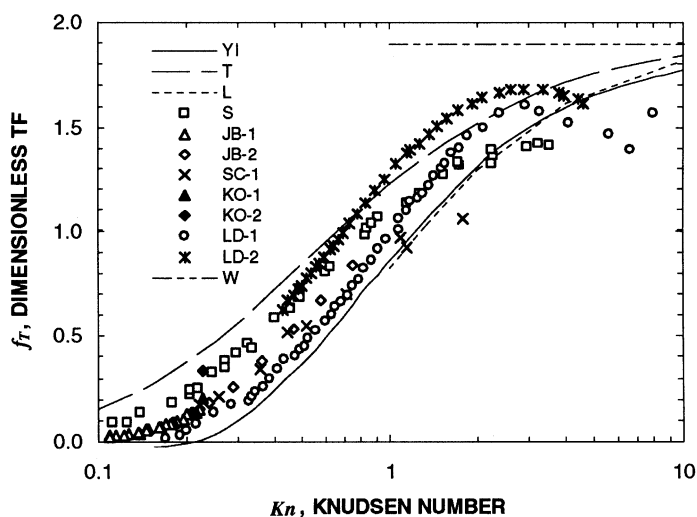


Fig. 3. Comparison of the theoretical solutions for $\kappa_{21} = 0.01$ with the thermophoretic force data for particles of high thermal conductivity. L: hard sphere gas, Loyalka [27]; YI: BGK model, Yamamoto and Ishihara [26]; T: interpolation formula, Talbot [31]; W: free molecule limit, Waldmann [8]; Data legends are listed in Table 4.

recommended by Talbot [31] are compared in Figs. 3 and 4 with the available data for thermophoretic force. The Knudsen number and the dimensionless thermophoretic force were all converted according to the definitions given by Eqs. (2), (3) and (11). All of the experimental data were taken with single-particle techniques using modified Millikan cells or electrodynamic balances, except for those by Keng and Orr [32], whose thermophoretic force measurements were based on the motion of aerosol clouds in a laminar flow system. The details of the measurement technique are reviewed in later sections. The types of the particle-gas systems included in the data are summarized in Table 4.

It can be seen that most of the experimental data points fall between the curve defined by the various solutions of the Boltzmann equation and that by Talbot's interpolation formula for both cases of high and low relative thermal conductivity of particle to gas. In general, the agreement between the data and the theories is reasonably good.

2.4. Effect of finite gas volume

All of the theories discussed above are based on the assumption that the particle is surrounded by an infinite expanse of gas. However, the gas is always enclosed by some boundaries separated by finite distance in experiments. If the pressure is low enough the gas mean free path may become comparable to this length scale and the gas can no longer be treated as a continuous medium near the system

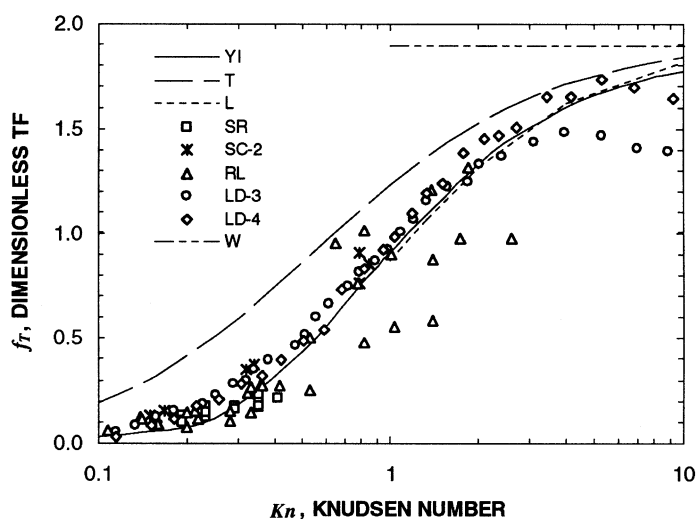


Fig. 4. Comparison of the theoretical solutions for $\kappa_{21} = 0.1$ with the thermophoretic force data for particles of low thermal conductivity. L: hard sphere gas, Loyalka [27]; YI: BGK model, Yamamoto and Ishihara [26]; T: interpolation formula, Talbot [31]; W: free molecule limit, Waldmann [8]; Data legends are listed in Table 4.

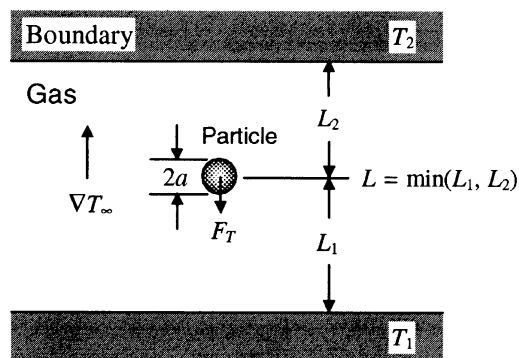


Fig. 5. Thermophoresis in finite volume of gas.

boundaries. Under such a condition an additional Knudsen number becomes relevant,

$$Kn_L = \frac{\ell}{L} \quad (16)$$

where L is the length scale of the gas environment. If the gas temperature gradient is established by keeping two parallel plates at different temperatures, as illustrated in Fig. 5, L may be defined as the minimum of the distances from the particle to the hot and the cold plates. Note that Kn_L is always smaller than Kn since the particle radius a must be smaller than L .

For sufficiently large Kn_L , there are temperature jumps at the hot and cold surfaces that must be taken into account to determine the temperature gradient in the apparatus. For $Kn \gg 1$ and $Kn_L \gg 1$, Brock [33] found the thermophoretic force is given by

$$F_T = \frac{\pi}{4} a^2 p \frac{T_2 - T_1}{T_0} \quad (17)$$

where T_2 and T_1 are the temperatures at the hot and cold plates, respectively, p and T_0 are the gas pressure and temperature at the location of the particle, respectively.

Phillips [12] found an approximate solution of the Boltzmann equation for arbitrary values of Kn_L with the restriction $a \ll L$. His result can be related to Waldmann's free molecule limit by

$$\frac{F_T}{F_{T, FM}} = \frac{\left[1 + \frac{1}{2}(\alpha_{mt} - \alpha_{mn})\right] \left[1 + \frac{15}{4} \left(\frac{45\pi}{128}\right) Kn_L\right]}{1 + \frac{15}{4} Kn_L + \left(\frac{15}{4}\right)^2 Kn_L^2 + \frac{15}{4} Kn_L \left(\frac{3}{2} + \frac{15}{4} Kn_L\right) (\alpha_1^{-1} + \alpha_2^{-1} - 2)} \quad (18)$$

Here α_{mt} and α_{mn} are the tangential and normal momentum accommodation coefficients of the particle, α_1 and α_2 are the thermal accommodation coefficients of the two plates, respectively.

For complete accommodation, the above equation reduces to Waldmann's free molecule result, Eq. (9), in the limit of $Kn_L \rightarrow 0$. In the limit of $Kn_L \rightarrow \infty$, it recovers Brock's solution, Eq. (17). Based on the Chapman–Enskog treatment of the Boltzmann equation, Annis and Mason [34] attempted to relax the restriction of $a \ll L$. However, their results in the limiting case of $a \ll L$ agree with Phillips' solution only when Kn_L is either very large or very small, which they thought was caused by their casual treatment of the gas thermal conductivity.

Quantitative measurements of the effect of finite gas volume are scarce. Tong [17] measured the thermophoretic force for three different spacing between the heat exchanger. Davis and Adair [16] compared their data in the free molecule regime with the Phillips' theory and found Eq. (18) agreed with the trend of their data. Li and Davis [13,14] made corrections to their thermophoretic force data at large Knudsen numbers for the effect of the temperature jumps at the heat exchanger plates.

3. Previous Results for Non-spherical Particles

Thermophoresis of non-spherical particles is more complex than the case of spheres. The theoretical solutions are more difficult and additional parameters are needed, such as the particle shape and its orientation in the temperature field. Our current knowledge of the thermophoresis of non-spherical particles is still very limited. There are a few asymptotic solutions for the thermophoretic force in the continuum and the free molecule limits for several simple particle shapes. No experimental data are available to test the validity of these theories.

3.1. Near-continuum regime

Many aerosol particles of practical interest can be approximated as cylinders, spheroids, simple two-sphere aggregates, or multiple-sphere aggregates of various structures, etc. Among the above shapes, cylinders are the simplest. Reed [35] did a theoretical investigation of the thermophoresis of cylindrical particles in the near-continuum regime. He found that the thermophoretic velocity of a cylindrical particle with its axis perpendicular to the temperature gradient is only approximately 55% of the sphere value.

Leong [36] and Williams [37] obtained analytical solutions in the near-continuum regime for the thermophoresis of spheroids with their axes of revolution parallel to the temperature gradient. In Fig. 6 the calculated thermophoretic force and velocity, normalized by the values for a sphere of the same volume as the spheroid, respectively, are plotted vs. the spheroid axis ratio, which is defined as the ratio of the lengths of the polar and equatorial axes. It can be seen that a prolate particle experiences a greater thermophoretic force than a sphere of the same volume,

whereas the force on an oblate particle is smaller than that on the equivalent sphere. Under the influence of the thermophoretic force, a prolate spheroid drifts faster than a sphere of the same volume, whereas an oblate spheroid drifts slower. For both prolate and oblate spheroids, the higher the particle thermal conductivity, the more the particle shape affects the thermophoretic force and velocity. Most particles of practical interest have a thermal conductivity at least 10 times greater than that of the surrounding gas, i.e. $\kappa_{21} \leq 0.1$. For $\kappa_{21} = 0.1$, the thermophoretic force on an oblate spheroid of axis ratio 0.5 is 56% of that on a sphere of the same volume and the thermophoretic velocity is 49% of the sphere value. Under the same conditions the thermophoretic force on a prolate spheroid of axis ratio 2.0 is 85% greater than that on the equivalent sphere and the thermophoretic velocity is 94% greater. Therefore, the effect of particle shape on thermophoresis predicted by the above theories is quite significant in the near-continuum regime.

Williams [38] was the first to analyze the thermophoresis of two-sphere aggregates. He obtained an approximation in the near-continuum regime for a cluster of two identical spheres joined in the direction of the temperature gradient. The thermophoretic force on the aggregate was found to be 7.9% greater than that on a single sphere of the same volume, whereas the thermophoretic velocity was found to be 5.4% greater than the sphere value.

Mackowski [39] used a method based on spherical-harmonics to analyze the thermophoresis of two-sphere aggregates of high thermal conductivity in the near-continuum regime. He found that the thermophoretic velocities of a symmetric aggregate particle differ by as much as 20% with the line of centers parallel or perpendicular to the temperature gradient. Consequently, an obliquely aligned

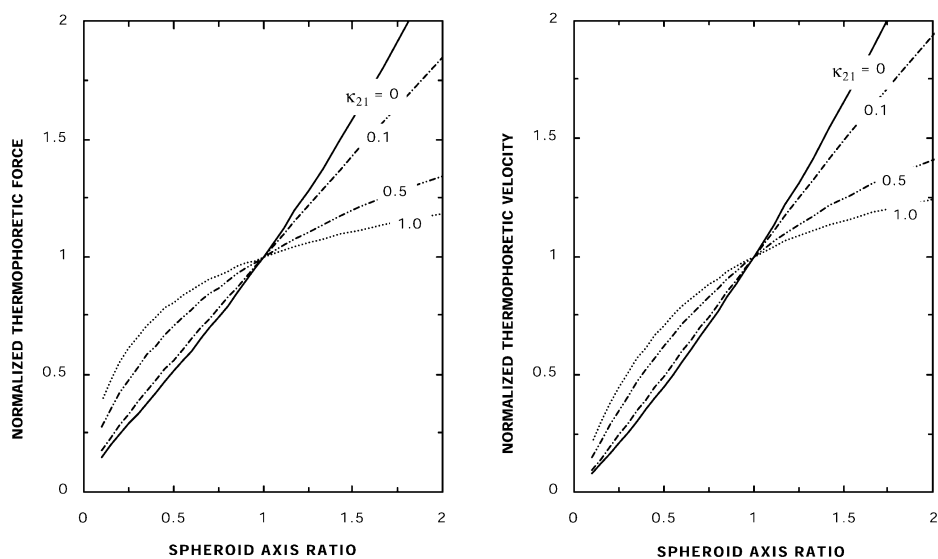


Fig. 6. Effects of the axis ratio on thermophoretic force and velocity for spheroids.

aggregate will not move in the same direction as the temperature gradient. For example, a symmetric aggregate particle ($Kn = 0.1$ and $\kappa_{21} = 0.05$) can drift in a direction up to 6.4° from the temperature gradient. Mackowski also found that the thermophoretic force on an asymmetric aggregate results in a net torque, which tends to rotate the aggregate so that the larger sphere is on the upstream (higher temperature) side.

Rosner et al. [40] discussed the effect of particle size and shape on thermophoresis. They attempted to extend Mackowski's method to multiple-sphere aggregates and showed some preliminary results on straight chains and randomly packed clusters in the near-continuum regime. They found that, in the limit of infinite chain length, the thermophoretic velocity of the aggregates parallel and perpendicular to the temperature gradient are approximately twice and two-thirds, respectively, of those of a single unit sphere. Averaged over all possible orientations, both the thermophoretic force and velocity are approximately 10% larger than the unit sphere values. For random clusters of large number of unit spheres the thermophoretic force and velocity are approximately 20% smaller than the unit sphere values.

3.2. Free molecule and transition regimes

Garcia-Ybarra and Rosner [41] analyzed the thermophoretic force on spherocylindrical particles in the free molecule regime, using the same method as Waldmann [8]. The particles have the shape of a cylinder but with two hemispheres of the same radius at both ends. They found that if a long particle of this type is aligned parallel to the temperature gradient, its thermophoretic velocity is 31% greater than the value of a sphere of the same radius as the cylinder, whereas if the same particle is aligned to be perpendicular to the gradient it will drift approximately 6% slower than an equivalent sphere. They also found that there is no net torque on such particles but the direction of the thermophoretic velocity can deviate from the temperature gradient by as much as 12° if the particle is aligned at an angle to the temperature gradient.

For aggregate of spherical particles, no theoretical solution for the free molecule limit is currently available. Unlike the spherocylinders treated by Garcia-Ybarra and Rosner, aggregates of spheres have concave surfaces with which the same gas molecules may have multiple collisions before vanishing into the space. This must be accounted for in the calculations. For other convex shapes in contrast, applying Waldmann's approach for the free molecule limits is straightforward.

No theoretical solution is currently available for the thermophoresis of nonspherical particles in the transition regime. However, we expect both the thermophoretic force and the velocity to be sensitive to particle shape and orientation as in the free molecule and near-continuum cases.

Experimental data showing the effect of particle shape and orientation are few. Davis and Adair [42] measured the thermophoretic force on disks of approximately 1 cm in diameter. Their experiments, however, are prone to the error caused by gas convection, because the temperature field was set up horizontally.

The latest measurements done by our group [54] provided data for the effect of particle shape on thermophoresis for aggregates of spheres for the first time. The thermophoretic force has been measured for polystyrene latex (PSL) spheres and aggregates of spheres in nitrogen and helium in the transition regime. The effect of particle shape was determined from the data in terms of a thermophoretic shape factor κ_T , which is defined as the ratio of the thermophoretic force on a non-spherical particle to that on a sphere of equal volume with other conditions being the same,

$$\kappa_T = F_{T, \text{ non-sphere}} / F_{T, \text{ equivalent volume sphere}} \quad (19)$$

It was found that the thermophoretic shape factors measured at one Knudsen number in the transition regime can be used to represent the shape effect over fairly large Knudsen number range. For PSL doublets of equal spheres aligned such that the line of centers is parallel to the temperature field, the measured thermophoretic shape factors are $\kappa_T = 1.13 \pm 0.08$ at $Kn = 0.98$ in nitrogen and $\kappa_T = 1.04 \pm 0.03$ at $Kn = 0.65$ in helium, which compare fairly well with the continuum limits predicted by Rosner et al. [40] and Williams [38]. For PSL triangular triplets of equal spheres aligned such that the plane of centers is parallel to the temperature field, the shape factors are $\kappa_T = 1.20 \pm 0.12$ at $Kn = 0.98$ in nitrogen and $\kappa_T = 1.06 \pm 0.06$ at $Kn = 0.65$ in helium. The shape factor was found to increase with the number of spheres in the aggregate particle, consistent with the trend predicted by the existing theories [40].

4. Review of measurement techniques

The measurement techniques for thermophoretic effect can be categorized by whether the measurement is made on single particles or on groups of particles. Single-particle measurements have many advantages over those based on a group of particles, the most important one being that individual particle size can be determined accurately. This is critical because thermophoresis depends strongly on particle size.

In the early years, the thickness of the particle-free regions around heated bodies like those observed by Tyndall [1] was measured [43,44]. It was recognized that quantitative measurements would not be successful until convection in the gas was minimized and the temperature gradient was controlled. Later, various flow systems with better control for the temperature gradient were developed [15,32,45]. In all of these systems multiple aerosol particles were injected into and carried by a laminar flow of gas, which was perpendicular to a temperature gradient. The thermophoretic velocity of the particles was inferred by observing the drift motion of the particle group in the temperature field directly or by measuring the location of particle deposition a certain distance downstream. The results of these measurements were difficult to interpret because of the uncertainty in the effect of convection and in the individual particle size.

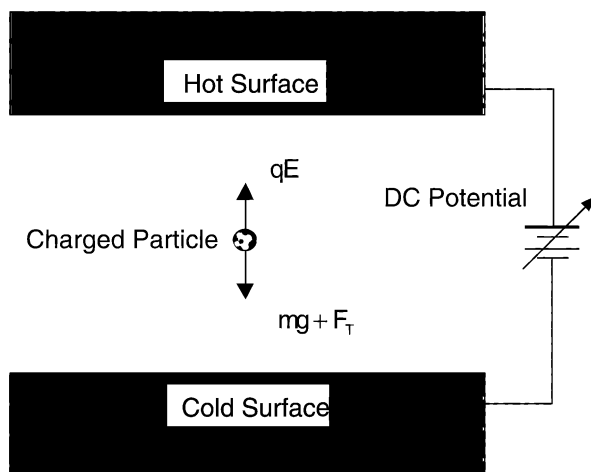


Fig. 7. Modified Millikan cell used in previous thermophoresis measurements.

In some later experiments the sedimentation of a group of particles in a temperature field was measured in stationary gases [46–48]. As in the previous experiments, the sizes of the particles were not determined individually. Some researchers [16,17,42] used small objects mechanically suspended in a stationary gas. Although it was not difficult to measure the particle size (in fact the particles had to be large enough for mechanical attachment), the temperature gradient had to be set up horizontally because of the vertical supporting rod or wire. This caused the serious problem of convection flow and large measurement error.

The thermophoresis measurement on single aerosol particles started with Rosenblatt and LaMer's experiments with a modified Millikan cell [21], as illustrated in Fig. 7. They levitated single charged liquid droplets between two parallel electrode surfaces by applying a dc potential across the electrodes. The top surface was heated and the bottom surface was cooled to set up a vertical temperature gradient that suppresses natural convection. The particle size was determined from its sedimentation speed in the gravitational field before the temperature gradient was established. This configuration was adopted by many later researchers [18,20,22,49].

In some experiments using modified Millikan cells the particle was allowed to free-fall in the temperature gradient and the thermophoretic velocity was measured, while in others the particle was kept stationary, and the ratio of the thermophoretic force to the particle weight was measured from the change in the dc potentials before and after the temperature gradient was established. Fuchs [50] noted that the thermophoretic force method is more useful than the thermophoretic velocity method because a moving particle in a temperature gradient will generally move into regions of different local temperature and this effect is not built into most of the existing theories.

The main drawback of a modified Millikan cell is that the levitated particle is not stable horizontally. Li and Davis [13,14] solved this problem by using an electrodynamic balance (EDB) in their thermophoretic force measurements. Like in a Millikan cell, a dc potential was applied across the top and the bottom electrodes to levitate a charged particle. In addition, an ac field was established to stabilize the particle in both the vertical and horizontal directions. An extensive review of the various configurations of EDB was given by Hartung and Avedisian [51]. Li and Davis mounted a double-ring EDB between two heat exchanger plates in a vacuum chamber, as shown in Fig. 8. With this configuration they were able to extend greatly the range of parameters and Knudsen numbers of thermophoretic force data.

One drawback of Li and Davis' apparatus is that although the particle is stabilized horizontally, there is no mechanism to neutralize any horizontal force on the particle. Such forces are frequently encountered due to imperfect alignment of the electrode rings, the gas flow due to convection or pumping, or simply the horizontal component of the thermophoretic force on a non-spherical particle. If these horizontal forces are not compensated, trapped particles will experience a horizontal oscillation, which is detrimental to both thermophoretic force and particle size measurements.

The latest development in thermophoresis measurements is the use of an electrodynamic balance capable of force compensation and measurement in three-

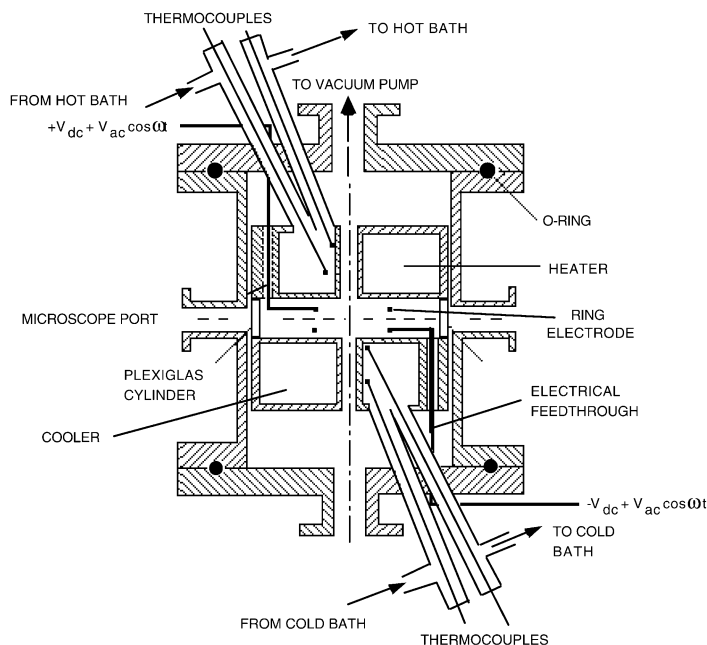


Fig. 8. A cross-section of the electrodynamic balance and vacuum chamber used by Li and Davis [13,14] for thermophoretic force measurements.

dimensions detailed by Zheng et al. [52]. The three-dimensional force compensation affords better accuracy in force measurements and is especially suited for non-spherical particle thermophoresis experiments since for these particles the direction of the thermophoretic force depends on particle orientation and is therefore three-dimensional in general.

Some of the recent experiments [47–49] were carried out under microgravity conditions in drop towers or during parabolic flights using the sedimentation cell technique mentioned above. The advantage of a microgravity environment is obvious. The elimination of convection in the gas improves the accuracy of thermophoretic force or velocity measurements. This is especially attractive for measurements at small Knudsen numbers, where the convection problem is the most serious. However, individual particle size was not measured in any of the above microgravity experiments, and therefore the advantage of microgravity was not used to its full strength.

5. Summary

The current state of thermophoresis study is that the theories for spherical particles are now available in the entire Knudsen number range, but for non-spherical particles only the near-continuum and free-molecule asymptotes have been worked out for a handful types of particle shapes. While a considerable amount of experimental data has been accumulated for the thermophoresis of spheres, the data for non-spherical particles are relatively few.

For spherical particles, the agreement between the theories and the data is reasonably good, but there is a lot of room for improvement. All of the recent solutions of the Boltzmann equation contain the assumption of complete accommodation at the particle surface, which is certainly not true under most experimental conditions. This causes the theoretical predictions to deviate from the experimental data, especially for gases such as helium. Therefore, it is hoped that the accommodation coefficients are built into the future solutions of the Boltzmann equation and that the solutions are provided in a form convenient for fitting the experimental data.

For thermophoretic force measurements, particle size must be determined accurately for individual particles. Otherwise, the uncertainty in particle size will become the major source of error in the thermophoretic force data. In this regard, measurements based on single particles are favored over those based on multiple particles. Other factors such as convection and the effect of finite gas volume must be carefully considered in planning out the experiments. Electrodynamic balance (EDB) provides the best platform for single particle size and force measurements while the use of a microgravity facility, such as drop towers and parabolic flights, eliminates convection. The combination of the above two techniques seems to be a very promising method of thermophoretic force measurements, which is especially useful in providing high quality data at small Knudsen numbers to elucidate

whether negative thermophoresis exists. In the previous microgravity-based experiments electrodynamic balance has not been used.

As mentioned above our knowledge of the non-spherical particle thermophoresis is limited. Theories applicable to intermediate Knudsen numbers are needed. Experimental data are needed for checking the existing near-continuum and free molecule theories. An experimental approach using EDB is promising, because individual particles can be retained for extended duration, which is necessary to characterize the particle geometry in detail. Experiments [53,54] have shown that non-spherical particles trapped in EDB typically have distinct orientations relative to the electric field. Therefore, thermophoretic force data for these particles using the EDB will have the advantage of no ambiguity as to the particle orientation relative to the temperature field. However, as the thermophoretic force on a non-spherical particle is three-dimensional in general, an EDB capable of three-dimensional force measurements should be used.

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